Weight and local gravity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish mass from gravitational weight.

SUCCESS CRITERIA - I can...

1. Use mass in kg and force in N.
2. Calculate weight using the stated local g.
3. Explain how weight changes while mass stays the same.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “weight” and “mass” to explain your answer. Sentence starters are optional.

weight: the gravitational force on an object

mass: a measure of how strongly an object resists acceleration

field: a description of an influence at each location

01 Warm-up: bring back an earlier idea

A simplified calculation predicts position (**-50**, **70**) km with bounds ±**5 km** in x and ±**7 km** in y. **Write a short briefing giving** both intervals and two reasons it is not a real asteroid collision forecast.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_g = mg$; $g = F_g/m$.
F_g is weight. Field strength in N/kg is numerically equal to free-fall acceleration in m/s².

Worked reasoning: A **3 kg** object in a field of **4 N/kg** has weight **3×4=12 N**. Moving it to a **2 N/kg** field halves weight to **6 N** but leaves mass **3 kg**.

Your turn: Draw weight arrows for the two places and **label the unchanged quantity**.

Sentence starter: In my work, I used … to show … . I labeled it … .

Location	Mass (kg)	g (N/kg)	Weight (N)
A	3	4	12
B	3	2	6

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The same **2 kg** object visits worlds with g=**10** and g=**5 N/kg**. **Compare its mass** and weight.

Sentence starter: Both … . They differ in … because … .

World	g (N/kg)
A	10
B	5

Marzano check - circle: ① ② ③

04 Apply the physics

An astronaut has mass **60 kg**. **Complete the location table** using W = mg. **Explain which property of the** astronaut stays the same.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Location	g (N/kg)	Weight (N)
Earth	9.8	____
Moon	1.6	____
Planet	____	240

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A scale reports **0 N** for an astronaut in an orbiting spacecraft. Does that prove Earth exerts no gravitational force? Explain.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** cart has **6 N** right and **38 N** left. **Find acceleration**. **What else** must you know to decide if it speeds up?
Draw the forces on the chosen object and **label their sources**.

Word help: acceleration = rate of change of velocity

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

An object weighs **49 N** where g=**9.8 N/kg**. **Find its mass** and weight where g=**2 N/kg**.

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Draw and label the gravitational force.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

An object weighs **72 N** at one location and **18 N** at another where g=**3 N/kg**. **Find its mass** and the first field strength.

Sentence starter: I found … [unit]. This value means … in this situation.

Case	Mass (kg)	g (N/kg)	Weight (N)
1	____	____	____
2	____	____	____

Marzano check - circle: 1 2 3